

Course Title: Optimization

Course Code: IE402

Credit Structure(L-T-P-Cr): 3-0-2-4

Prerequisites: Fundamental understanding of Linear Algebra, Matrix Theory.

Suggested Textbook/references: Operations Research by HA Taha 8th/9th Edition

Linear Programming by Howard Karloff

Combinatorial Optimisation by C. H. PAPADIMITRIOU
K. STEIGLITZ

Evaluation Scheme

In Semester Exams (30%)

End Semester Exam (50%)

Assignments and Homework (20%)

Course Outcome:

After the completion of course:

The students will be able to understand the application and working knowledge of optimization in selected areas.

The students will develop an insight and understanding with respect to formulation of optimization problems and designing the optimal solutions with applications in various domain and engineering applications.

The students will get acquainted with applications of optimization problems for some practical system model and few relevant work being carried out.

Lecture Plan/Contents:

Introduction to optimization methods [4 HOURS]

Introduction

Formulation of an LPP

Graphical Method to solve an LPP

Nature of a solution

Linear programming problem [10 HOURS]

Algebraic method

Simplex method

Revised simplex method

Duality

Dual simplex method

Sensitivity analysis

Transportation Problem [6 HOURS]

North-west corner Rule

Row-minimum method

Vogel's Approximation

u-v method

Integer Programming

NP-hardness

Totally Unimodular Matrix

Heuristic Algorithm - Cutting Plane Method

Game Theory [6 HOURS] •

- Introduction to Game Theory
- Game with pure and mixed strategies
- $2 \times n$ and $m \times 2$ Games
- LPP Formulation of $m \times n$ Games

Numerical Solution of Convex Optimization [8 HOURS]

- Gradient Descent, Steepest Descent
- Newton's Method
- Conjugate Direction and Conjugate Gradient Methods